

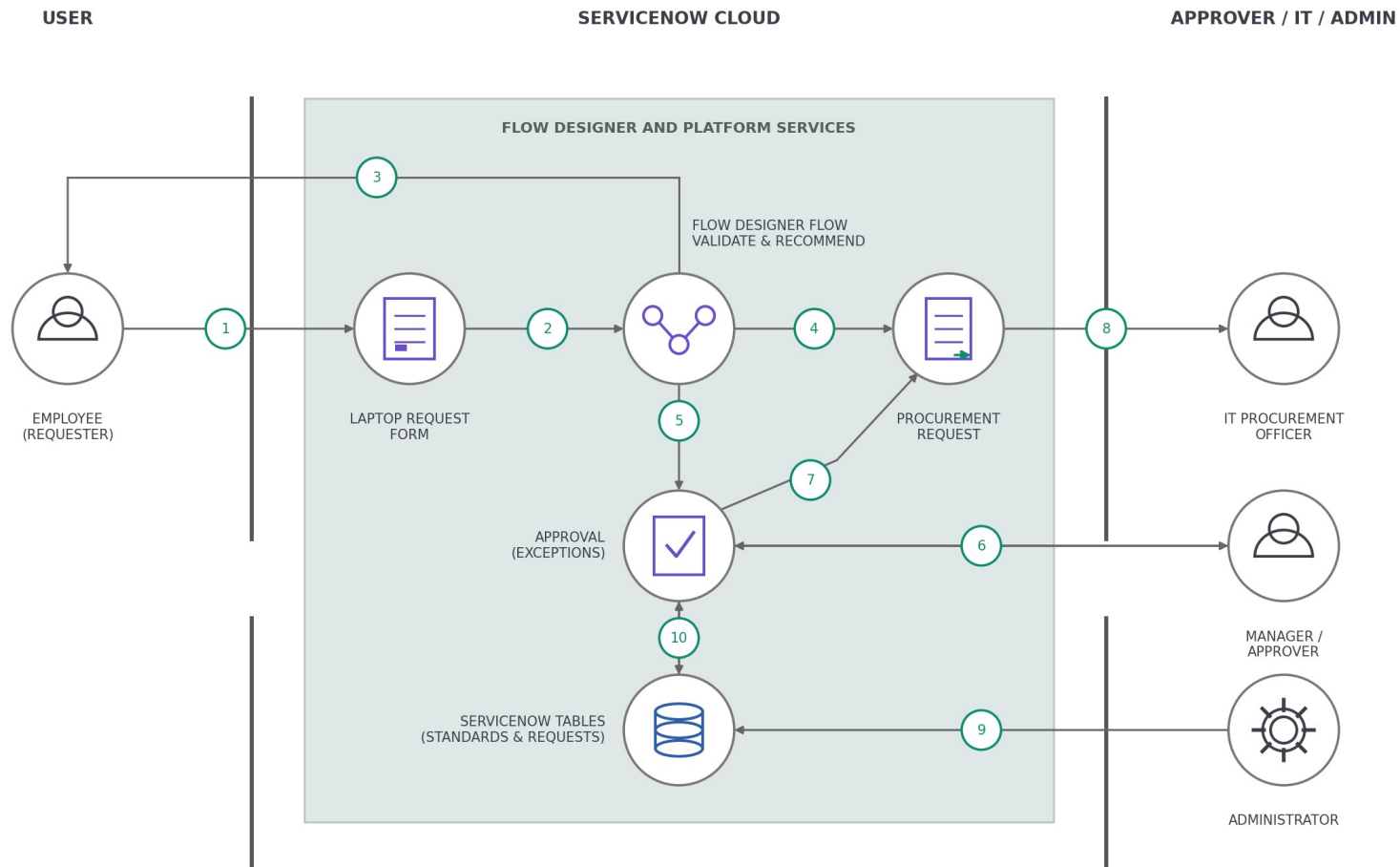
**Project Design Phase-II
Technology Stack (Architecture & Stack)**

Date	02 October 2026
Team ID	SWTID-2026-4912
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture: Standard laptop ordering with Flow Designer



1. The employee submits a laptop request (job role, required software and budget) through the laptop request form.
2. The request details are passed to the Flow Designer flow, which validates the requirements and matches them with the approved laptop configurations.
3. The recommended standard configuration and the request status are returned to the employee.
4. For standard requirements, the flow generates the procurement request.
5. Non-standard requirements are detected and routed to the approval step.

6. The approval request is sent to the manager / approver, and the decision is returned.
7. An approved exception continues to procurement request generation.
8. The procurement request is delivered to the IT procurement officer.
9. The administrator maintains job role requirement profiles and approved laptop configurations.
10. The flow reads the standards and stores requests, approvals and exception records in the ServiceNow tables.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Laptop request form where employees enter job role, required software and budget, view the recommendation and track request status; managers act on approval requests	ServiceNow Service Catalog item / Service Portal
2	Application Logic-1	Validation of request details and detection of non-standard requirements	Flow Designer (flow logic and conditions)
3	Application Logic-2	Recommendation of a standard laptop by matching job role, software and budget with approved CPU, RAM, storage and GPU configurations	Flow Designer (record lookup and decision logic)
4	Application Logic-3	Exception handling and approval workflow for non-standard requirements, with notifications to employee and IT procurement team	Flow Designer (approval and notification actions)
5	Application Logic-4	Automatic generation of the procurement request from the confirmed or approved configuration	Flow Designer (create record action)
6	Database	Job role requirement profiles, approved laptop configurations, laptop requests, approvals and exception records	ServiceNow platform tables
7	Cloud Database	Database service on cloud	ServiceNow cloud-hosted instance database (platform managed)
8	File Storage	Supporting documents attached to requests or exception justifications, if required	ServiceNow attachments
9	External API	Not used in the current design	Not applicable

10	Machine Learning Model	Not used; the recommendation is rule-based, matching requirements with approved configurations	Not applicable (rule-based logic in Flow Designer)
11	Infrastructure (Server / Cloud)	Application deployment on cloud Local Server Configuration: Not applicable Cloud Server Configuration: ServiceNow cloud-hosted instance	ServiceNow cloud platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source frameworks are required; the solution is built with the platform's low-code tools	Not applicable
2	Security Implementations	Role-based access for employee, manager / approver, IT procurement officer and administrator; only authorised users can approve exceptions or change standard configurations	ServiceNow roles and Access Control Lists (ACLs)
3	Scalable Architecture	Data-driven design: job role profiles and approved configurations are stored as records, so new roles and configurations are added without changing the flow	ServiceNow tables and Flow Designer
4	Availability	Hosted on a cloud platform, so availability of the application is managed by the platform	ServiceNow cloud platform
5	Performance	Requests are validated and matched automatically against a defined set of approved configuration records, with no manual steps	Flow Designer and ServiceNow tables

References:

<https://c4model.com/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>